

Leela Krishna Koppolu

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SUMMARY

AI Engineer with 4+ years of experience building AI/ML and GenAI applications and the data infrastructure behind them across banking, retail, and enterprise. Skilled in LLM-powered systems, RAG pipelines, vector search, feature engineering, and model serving, with deep AWS expertise (SageMaker, Bedrock, S3, Glue, Lambda, EMR, Redshift, Kinesis, EKS) and modern data tooling (Snowflake, Databricks, Spark/PySpark, Airflow, Kafka, dbt). Experienced in scalable ML/AI workflows, embeddings and retrieval systems, data-quality engineering, and Spark/SQL tuning that reduces latency and cost.

EDUCATION

Master of Science, Computer Science

Jan 2023 – Dec 2024

The University of Texas at Arlington, TX, USA

SKILLS

AI & ML Engineering:

LLMs, RAG Pipelines, Prompt Engineering, Embeddings, Semantic / Vector Search, LangChain, Hugging Face Transformers, PyTorch, scikit-learn, MLflow, Feature Store, Model Serving & Inference, Vector Databases (Pinecone, FAISS)

Cloud & ML Platforms:

AWS (SageMaker, Bedrock, S3, Lambda, Glue, EMR, Redshift, Kinesis, Step Functions, EKS, CloudWatch), Azure (Databricks, Data Factory, Synapse), Databricks, Snowflake, Delta Lake, Unity Catalog

Data Engineering:

Apache Spark, PySpark, Spark SQL, Apache Airflow, Apache Kafka, dbt, AWS Glue, Azure Data Factory, Debezium CDC, AWS DMS, Great Expectations

Languages & Frameworks:

Python, SQL, TypeScript, FastAPI, Pandas, Pydantic

Warehousing & Modeling:

Snowflake, Redshift, Star/Snowflake Schemas, SCD Type 1 & 2, Conformed Dimensions, Fact-Grain Design, Materialized Views

MLOps, DevOps & BI:

Docker, Kubernetes (EKS), Terraform, GitHub Actions, Jenkins, Azure DevOps, Prometheus, Grafana, OpenLineage, PagerDuty, Power BI (DAX, RLS), Tableau, PostgreSQL, Oracle, SQL Server, MySQL

CERTIFICATIONS

AWS Certified Data Engineer – Associate | Databricks Certified Data Engineer Associate | SnowPro Core | dbt Fundamentals

PROFESSIONAL EXPERIENCE

Keurig Dr Pepper Inc., Dallas, TX

Oct 2024 – Present

AI Engineer

- Building AI and GenAI applications on AWS powering ML use cases across sales, supply chain, manufacturing, and consumer analytics.
- Designing and deploying RAG pipelines and LLM-powered applications, integrating Snowflake, S3, and vector databases (Pinecone, FAISS) with embeddings from Amazon Bedrock and Hugging Face models.
- Building feature pipelines and a centralized feature store on Databricks for batch and real-time ML inference, ensuring training-serving consistency with point-in-time correct joins.
- Developing scalable ETL and feature-engineering workflows using PySpark on Databricks, AWS Glue, and Snowflake to deliver reliable, ML-ready datasets at scale.
- Orchestrating ML training and ELT workflows with Apache Airflow, integrating S3, Kafka, microservices, and partner APIs into curated lakehouse zones with full lineage in Unity Catalog.
- Tuning model-serving and Spark/Snowflake workloads through partition tuning, broadcast joins, clustering keys, and materialized views, driving substantial latency and cost reductions.
- Collaborating with data scientists, analysts, and business stakeholders to translate ambiguous requirements into trusted, observable AI and data products.

Deloitte, USA

Jun 2020 – Dec 2022

Software Engineer

- Designed and built end-to-end AWS ETL pipelines using S3, Glue, Lambda, Step Functions, and Redshift to move client data from mainframe, Oracle, and SQL Server systems into analytics-ready datasets for enterprise reporting and regulatory submissions.
- Implemented Apache Airflow as the orchestration backbone for hundreds of DAGs, introducing standardized patterns, exponential-backoff retries, PagerDuty alerting, and SLA dashboards that materially improved pipeline reliability.
- Tuned PySpark and Snowflake workloads through partition tuning, broadcast hints, clustering keys, and materialized views, cutting runtime and warehouse cost on the most expensive nightly jobs.
- Designed Redshift and Snowflake warehouse models (star schemas, SCD Type 1/2, conformed and late-arriving dimensions) supporting enterprise reporting and self-service BI for hundreds of analysts.
- Built customer-segmentation and propensity datasets (next-best-action, churn risk) enabling targeted campaigns and measurable uplift on retention and cross-sell metrics.